Airplane Scatter Plugin

Quick Reference · FM-DX-Webserver · Author: Highpoint

v2.3b · April 2026

✈ WHAT IS AIRPLANE SCATTER?

An aircraft at **8 000–12 000 m cruise altitude** reflects FM signals between a distant transmitter and your receiver — even when both are beyond each other's radio horizon.

The plugin **predicts these events in real time**, up to 3 min ahead, and displays them on an interactive map with a live countdown.

Best path lengths: **400–1 500 km** · Typical event duration: **15–60 s**

⚙ HOW IT WORKS

- TX database served from **tx_search.js RAM** (no redundant download) 2.1
- Optional **userlist1.csv** (fmscan.org) enriches TX data with beam directions, PS, PI 2.2
- Fetches live ADS-B positions every **15 seconds**
  adsb.one → adsb.lol → adsb.fi (auto-failover)
- Evaluates scatter geometry for every **aircraft × TX pair**
- Applies radio-horizon, ellipse, elevation-angle & **beam direction** gates 2.2
- **Two-Pass Topography Evaluation** to exclude terrain-blocked signals 2.3b
- Shows top candidates with **ETA countdown** on map
- All external API calls routed via **local Node proxy**

📊 SCATTER SCORE (0–100 %)

Factor	Weight
Sweet Spot (elevation angles) (TX σ=1.2°, RX σ=3.0°)	40 pts
Horizon margin (σ ≈ 4 000 m above envelope)	20 pts
Cross-track distance (σ = 25 km)	20 pts
Reflection angle (optimum 40°, σ = 25°)	10 pts
Fuselage alignment (⊥ bisector)	5 pts
TX ERP (logarithmic)	3 pts
FM frequency (87.5 vs 108 MHz)	2 pts

× Size multiplier: A5 heavy **×1.30** · A4 large ×1.15 · A3 ×1.00 · A2 ×0.85 · A1 ×0.70

× **Beam multiplier** 2.2 : score reduced to 0 if TX antenna beam does not cover the aircraft position or the RX direction (from userlist1.csv beam data)

Red ≥ 80 % Excellent **Orange ≥ 60 %** High
Yellow ≥ 40 % Medium **Green < 40 %** Low

⏱ ETA COUNTDOWN

Candidates appear **3 min before** and remain **1 min after** the crossing point.

−2:45 Approaching (blue)

NOW ✓ At crossing ±5 s (green)

+0:38 Past crossing (red)

Dead reckoning fills the gap between 15-second ADS-B updates — icon & ETA refresh *every second* smoothly.

Score forecast evaluates **5 time steps** (0, 60, 120, 180, 300 s ahead); best score is used. Early-exit at score ≥ 95.

📡 DATA SOURCES

- **TX DB:** tx_search.js RAM
  freq, ERP, city, country

🚧 GEOMETRY GATES

- TX–RX ≥ **400 km**

🗺 MAP PANEL

- 🔵 Receiver QTH dot

- **userlist1.csv** 2.2 beam, PS, PI, pol — from fmscan.org (login required)
- **ADS-B:** adsb.one / .lol / .fi
- **Elevation:** opentopodata.org SRTM 90 m, batch, cached
- **Elevation Cache:** local JSON file 2.3b
- **Audio streams:** api.fmlist.org
- **Flags:** flagcdn.com (ITU→ISO)
- **Proxy:** local Node server proxy

- Elevation angle to AC < **9°** (TX & RX)
- Both ends inside aircraft's **radio horizon**
- d_tx + d_rx ≤ **1.02–1.08 × d_direct**
- Along-track ≤ d_txrx × 1.1
- Aircraft > 50 kts & > 1 000 ft
- **Aircraft not blocked by mountains** 2.3b
- **TX beam covers RX direction** 2.2

- 🟡🟠🔴 TX dots (sized by ERP)
- ✈ Aircraft icon or Photo (colour = score)
- 📍 Additional sweet spot marker
- - - - Scatter path TX ↔ AC ↔ RX
- Left: candidate list sorted by |ETA|
- TX detail: **PS, PI code, polarisation** from userlist1.csv
- Bottom: terrain profile (interactive, Sweet Spot band)
- ▶ Audio stream Play/Stop per freq
- Status bar: AC count · candidates · DB size

🆕 NEW & CHANGED IN V2.3B

- **Local Elevation Cache** 2.3b — Automatically maintains a server-side `elevation_cache.json` file to locally store topographic points, significantly lowering external API queries and speeding up elevation resolutions.
- **Topographic Pre-Filtering** 2.3b — A new two-pass terrain validation algorithm ensures that aircraft situated physically below the horizon line due to real mountain terrain are instantly excluded during the scoring calculation. This drastically reduces false positives from terrain-obstructed paths.
- **Airplane Photos & Markers** 2.3a — Display of airplane photos on the map, plus Sweet Spot map markers.
- **Auto-move Webserver** 2.3a — Added function to automatically move the web server interface to the right side (idea by bojcha).

💡 TIPS FOR BEST RESULTS

- **Install userlist1.csv** for beam-aware scoring — download from fmscan.org (login required), select *FM+ (Tropo)* → *CSV format* → semicolon separator, download as `userlist1.csv` and place it in `plugins/AirplaneScatter/`.
- **Sweet Spot band** — if the aircraft dot is not inside the green band in the elevation profile, scatter geometry is suboptimal even with a high score.
- **Hover tooltip** — move the mouse over the elevation profile canvas to read terrain height and elevation angles at any path point.
- **Audio Streaming** — start the web radio ~30 s before crossing; streams are typically 15–45 s delayed. Compare audio burst with the stream to positively ID the station.
- Heavy jets (A380, B747) = **+30 % score bonus** — prioritise!
- Click ↺ **Reload** to force a fresh TX database download and clear caches

📥 HOW TO OBTAIN AND INSTALL USERLIST1.CSV (FMSCAN.ORG)

1. Register and log in at fmlist.org (free account required).
2. Navigate to **FMSCAN** → **Userlist** or go directly to fmscan.org/perseus.php?userlistmode=fm.
3. Select: *FM+ (Tropo)* · CSV format · **Semicolon** separator · Click **DOWNLOAD userlist1.csv** (left-click, wait ~1 minute).
4. Save the downloaded file as `userlist1.csv` and copy it into the `plugins/AirplaneScatter/` folder on your FM-DX-Webserver.
5. Restart the webserver and reload the plugin (↺ button). The plugin will load the file automatically and log the number of entries found.